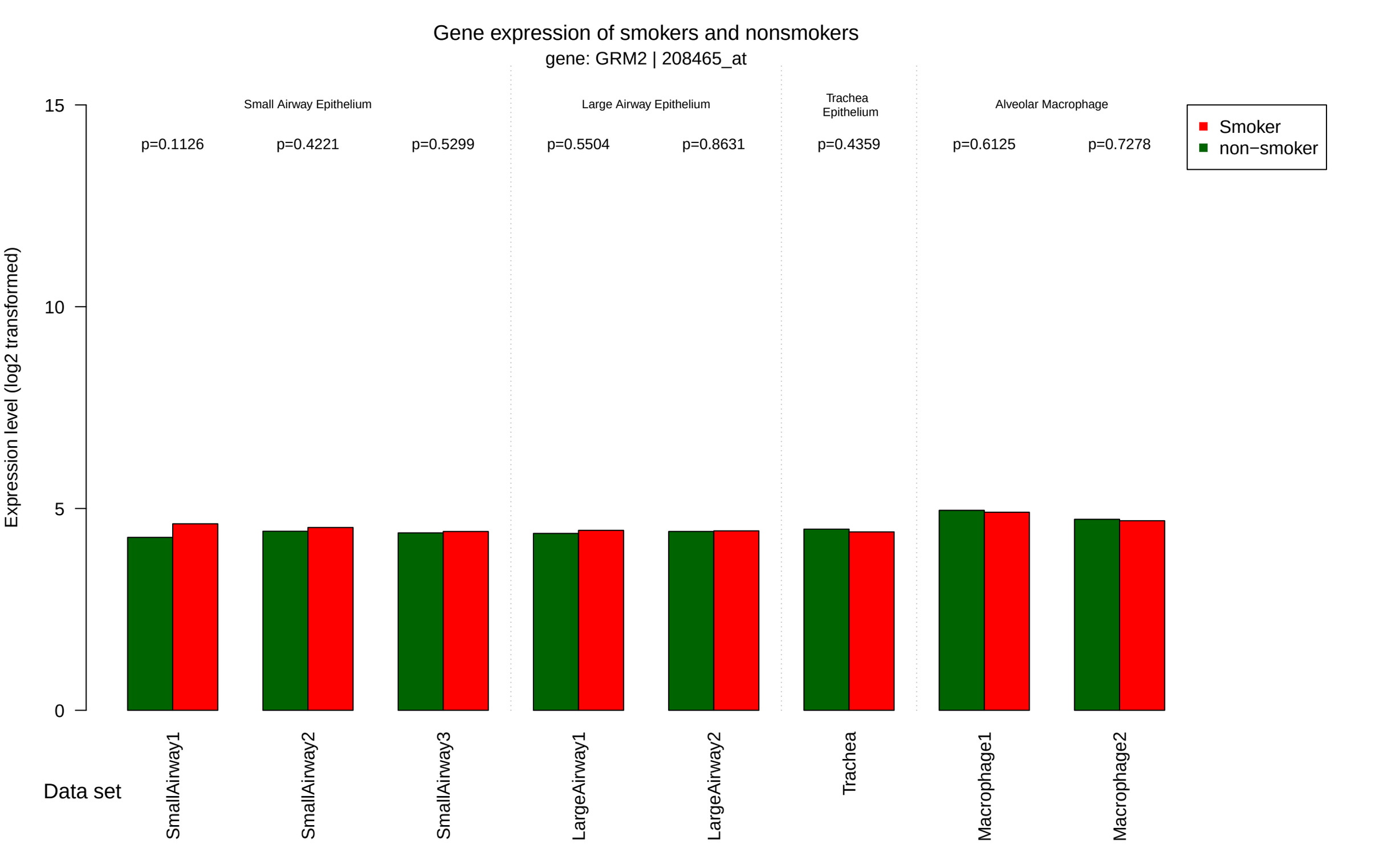
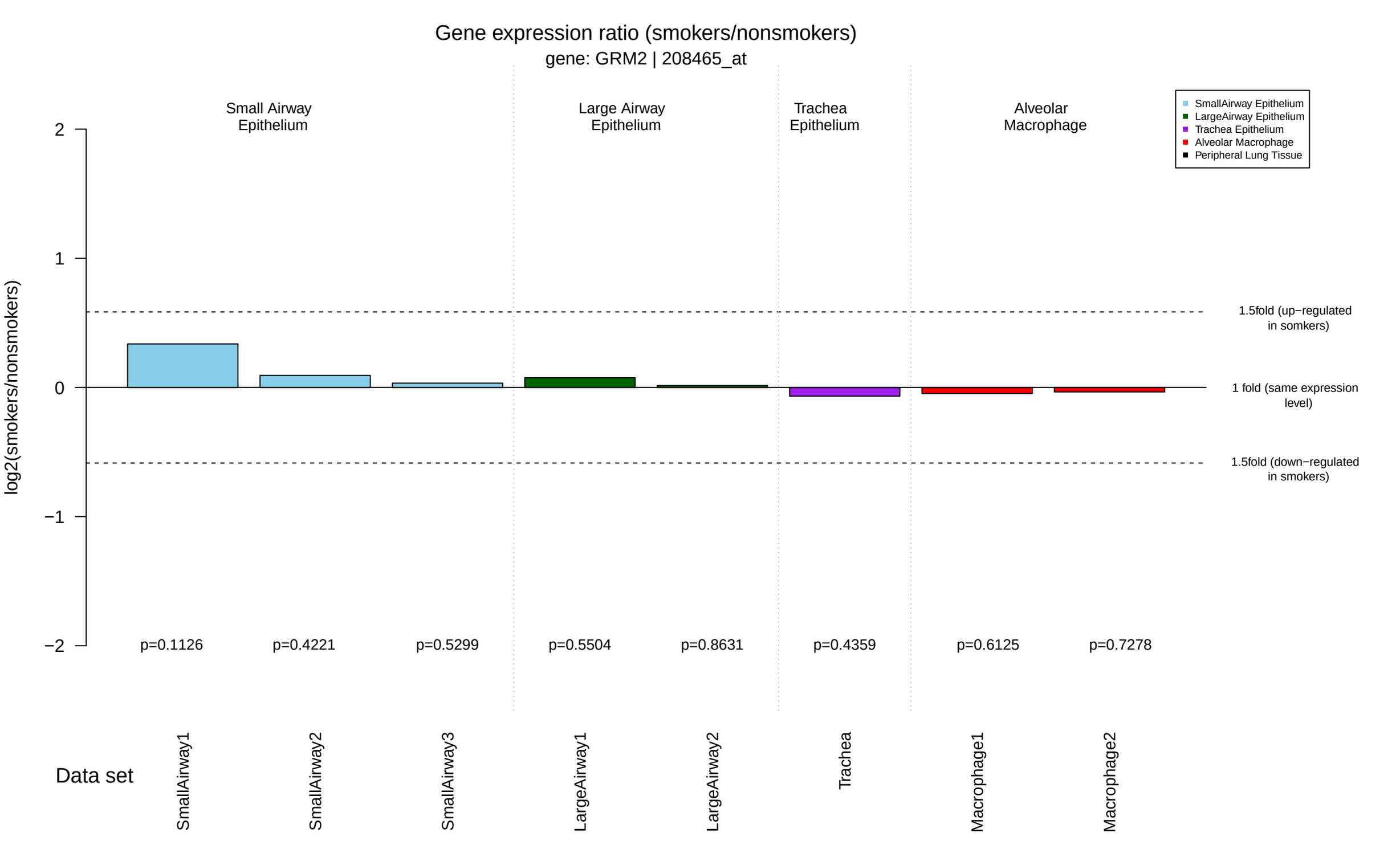
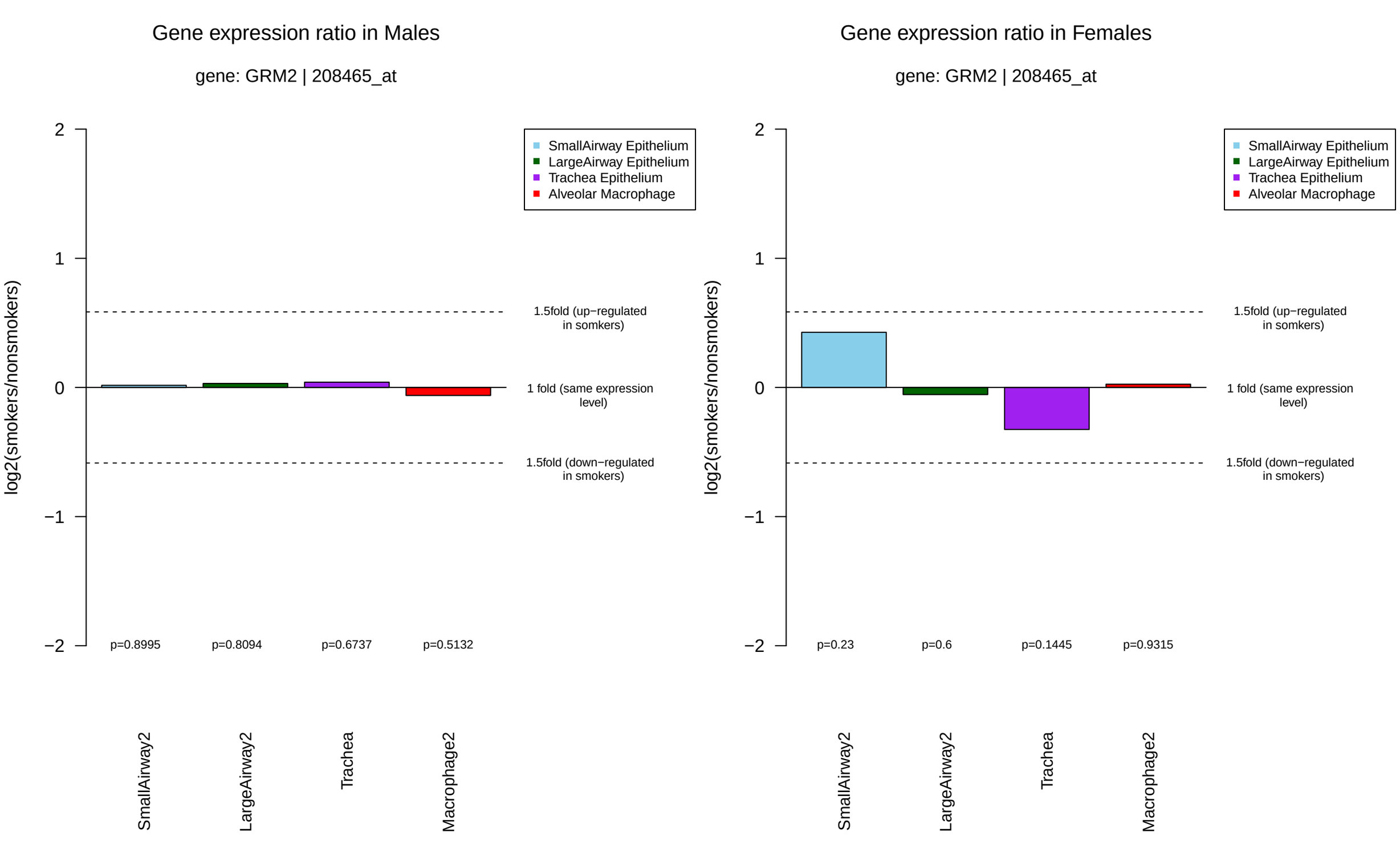
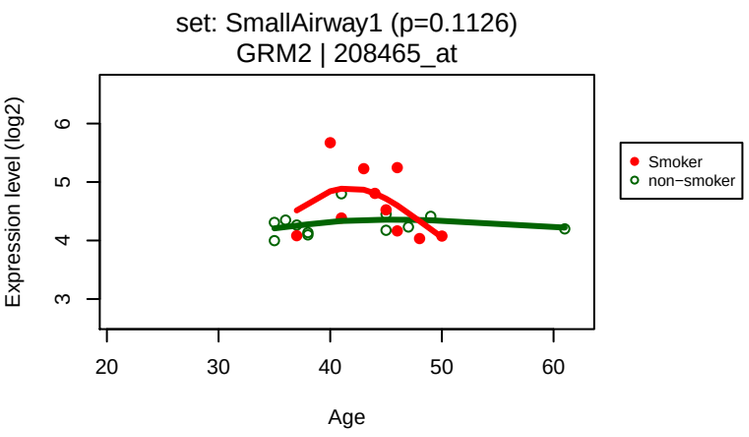
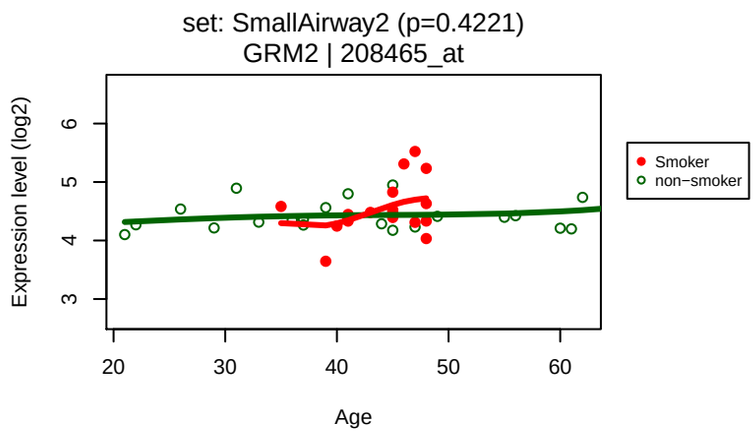
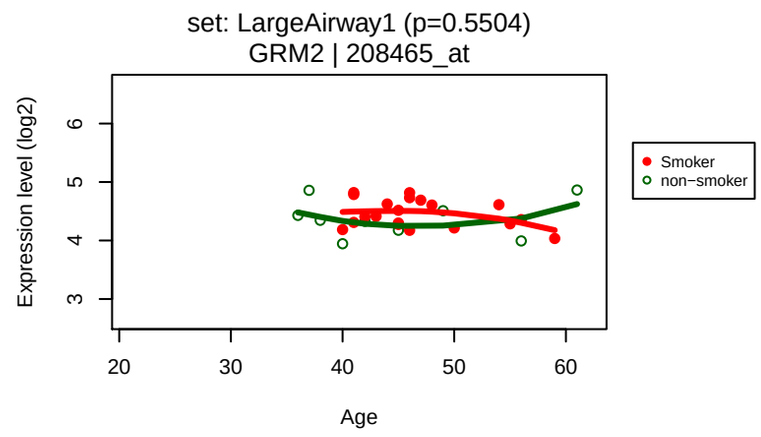

Gene expression of smokers and nonsmokers
gene: GRM2 | 208465_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1126
p=0.4221
p=0.5299
p=0.5504
p=0.8631
p=0.4359
p=0.6125
p=0.7278
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

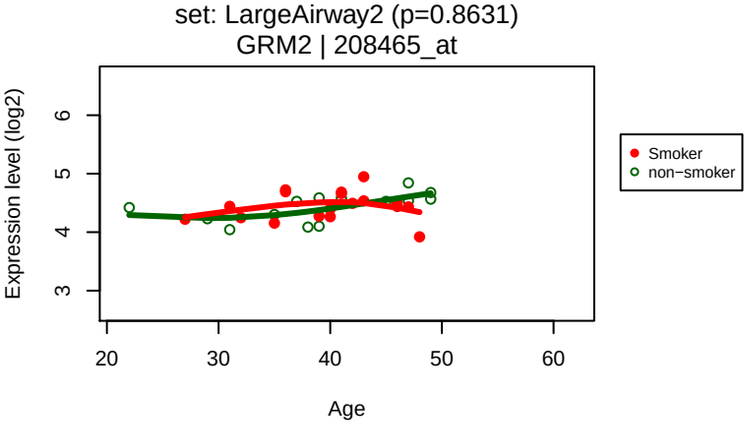

Gene expression ratio (smokers/nonsmokers)
gene: GRM2 | 208465_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1126
p=0.4221
p=0.5299
p=0.5504
p=0.8631
p=0.4359
p=0.6125
p=0.7278
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

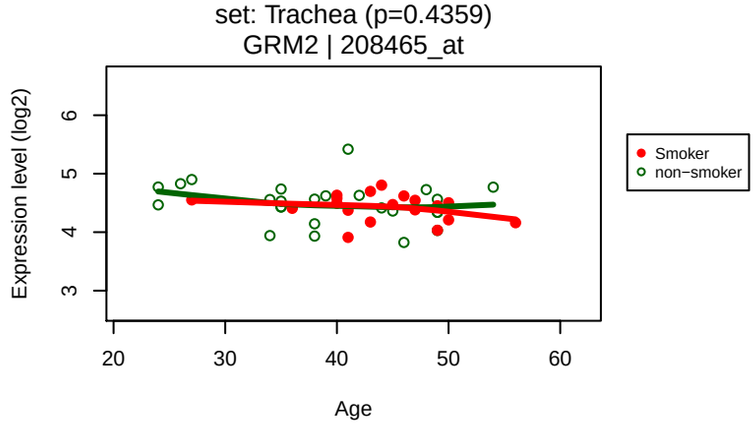

Gene expression ratio in Males
gene: GRM2 | 208465_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8995
p=0.8094
p=0.6737
p=0.5132
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRM2 | 208465_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.23
p=0.6
p=0.1445
p=0.9315
SmallAirway2
LargeAirway2
Trachea
Macrophage2

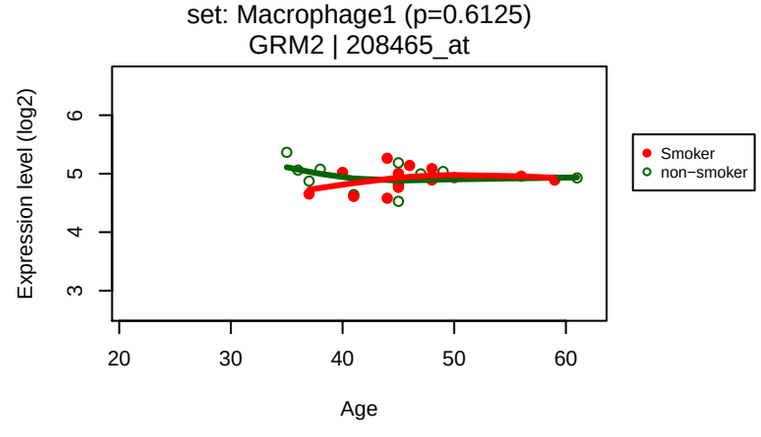

set: SmallAirway1 (p=0.1126)
GRM2 | 208465_at
Expression level (log2)
Age
Smoker
non−smoker

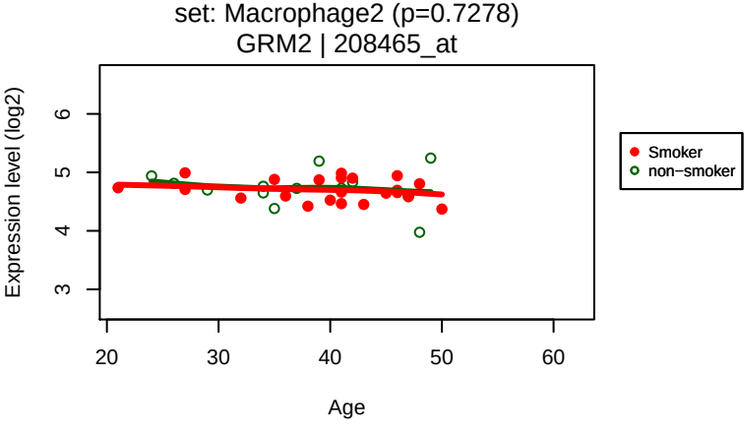

set: SmallAirway2 (p=0.4221)
GRM2 | 208465_at
Expression level (log2)
Age
Smoker
non−smoker

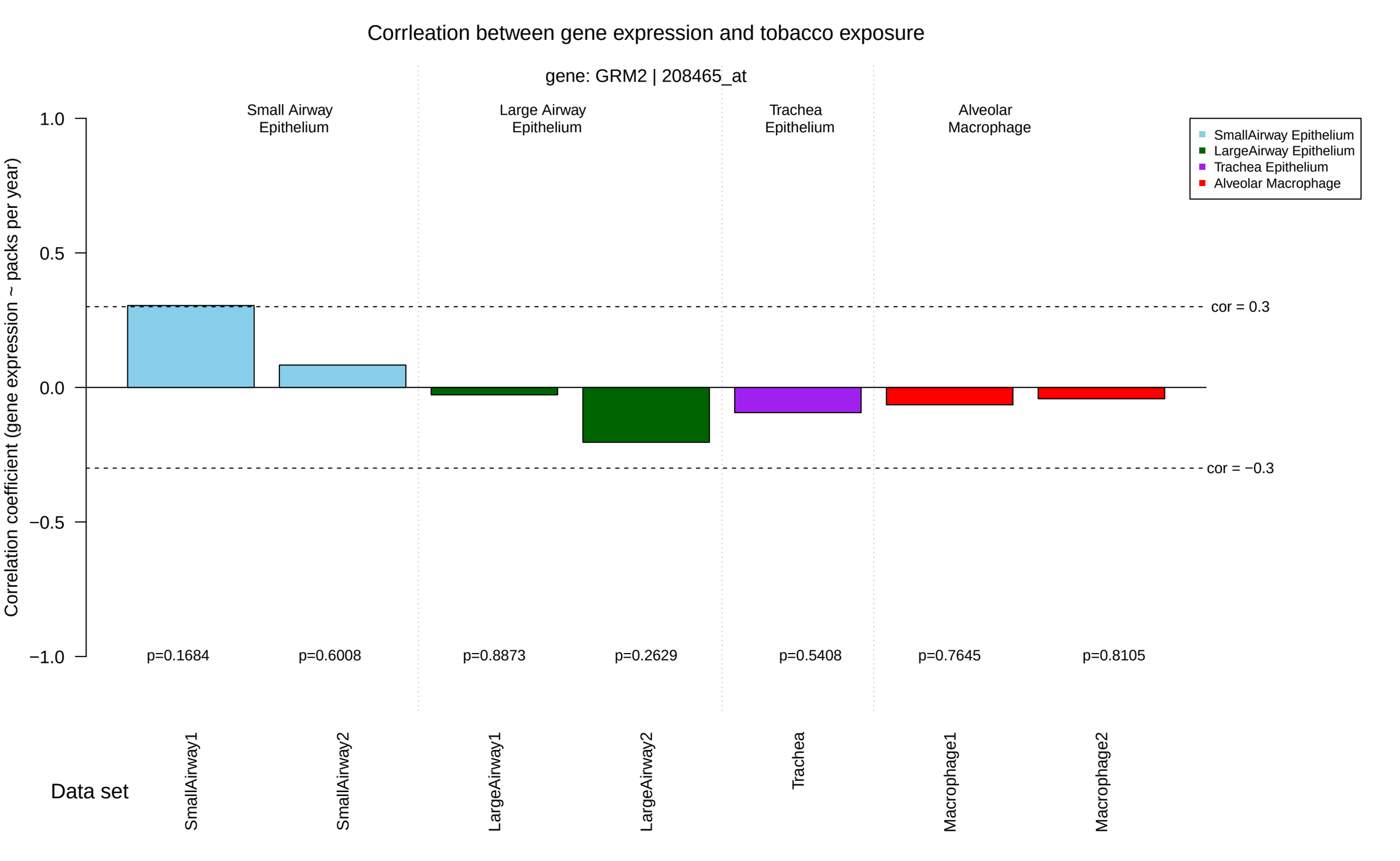

set: LargeAirway1 (p=0.5504)
GRM2 | 208465_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8631)
GRM2 | 208465_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4359)
GRM2 | 208465_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.6125)
GRM2 | 208465_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7278)
GRM2 | 208465_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRM2 | 208465_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1684
p=0.6008
p=0.8873
p=0.2629
p=0.5408
p=0.7645
p=0.8105
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2